



Asset Pricing with Deep Learning

Replication, Extensions and Uncertainty

Architecture, rolling performance and economic regimes

Reza Zamani, Hrafnhildur Lif Jonsdottir

Paraj Goyal and Elouan Bahri

Professor: Ali Kakhbod

UC Berkeley

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Presentation overview

The original paper

Economic pricing restrictions, hidden macro states and adversarial instruments.

Our replication

Published benchmarks, local model rankings and validation checks.

New results from Codes 08–12

Architecture, rolling windows, regimes, combinations and uncertainty.

The original paper's research question

Estimate a pricing kernel

Learn the stochastic discount factor that prices assets consistently with no arbitrage.

Use a richer information set

Firm characteristics and macroeconomic history allow risk exposures to vary over time.

Separate pricing from return prediction

An economic moment objective guides the network toward pricing-relevant information.

How the model learns asset prices

1. Summarize the economic environment

LSTM networks read macroeconomic history and create hidden states for the pricing and adversarial branches.

2. Build the pricing portfolio

A feedforward network combines firm characteristics with the pricing states to assign stock weights and construct the SDF.

3. Challenge and improve the model

The adversary finds instruments that reveal large pricing errors. The pricing network learns to reduce those errors.

The original model architecture

Component	Input and role	Output
Macro SDF encoder	178 macro series, LSTM	4 states
Pricing-weight network	46 characteristics + states	Two layers of 64 units
Macro instrument encoder	178 macro series, LSTM	32 states
Conditional network	Characteristics + states	8 bounded instruments

The pricing objective links the SDF network and the adversarial instruments.

Data and chronological evaluation

Sample	Period	Months
Training	1967–1986	240
Validation	1987–1991	60
Test	1992–2016	300

46 firm characteristics and 178 macroeconomic inputs

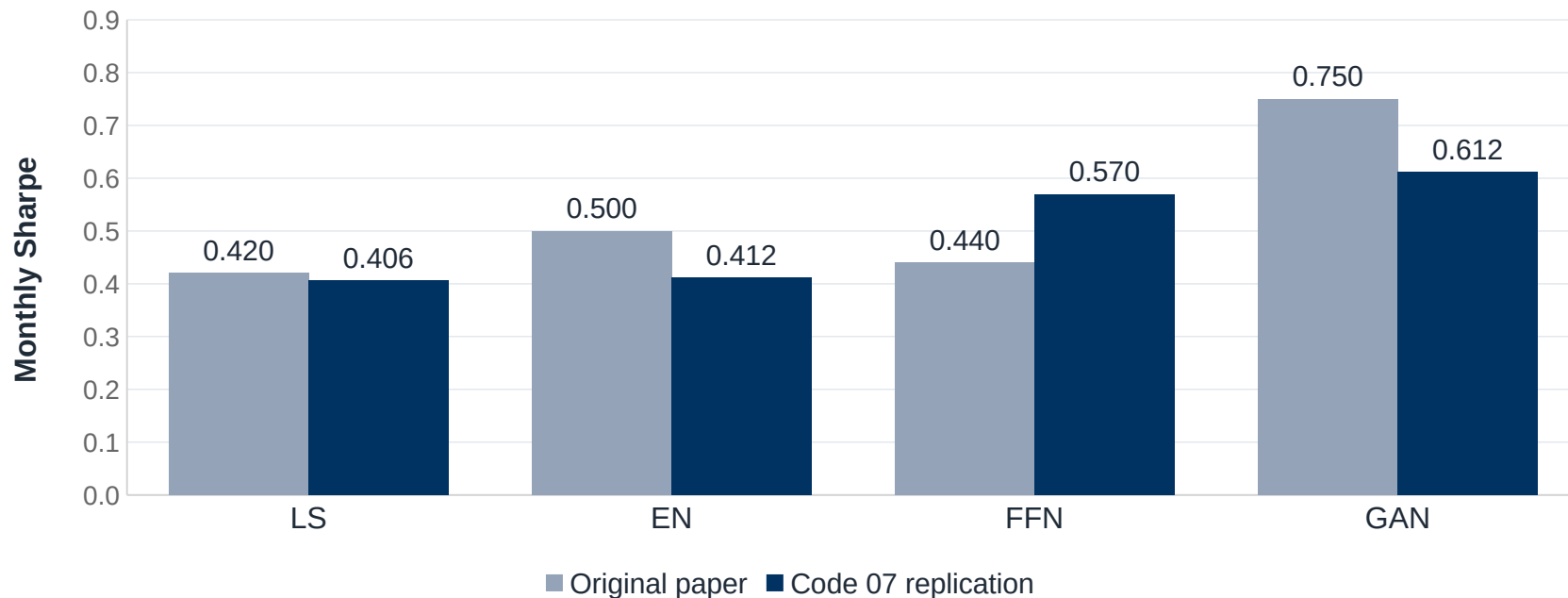
Fit on training months and select checkpoints using validation months.

The paper's published results

Model	Train	Validation	Test
LS	1.800	0.580	0.420
EN	1.370	1.150	0.500
FFN	0.450	0.420	0.440
GAN	2.680	1.430	0.750

GAN has the highest published monthly Sharpe in every split.

Replication versus the original paper



GAN stays first, while FFN and EN exchange test-period ranks.

Replication validation and common signals

Check	Our result	Reference
FF-3 test Sharpe	0.196	Paper 0.190
FF-5 test Sharpe	0.223	Paper 0.220
GAN test correlation	0.800	Authors' monthly GAN
Leading characteristics	ST_REV, SUV, r12_2	Same top three

These checks support the replication's economic similarity to the original.

Questions left by the previous presentations

Architecture

Would causal attention change the learned macro representation?

Time variation

Does one full-test Sharpe hide changes in relative performance?

Economic conditions

Do recession and volatility groups explain where the models differ?

What Codes 08–12 add

Code	Analysis	Question
08	LSTM versus Transformer	Which encoder works at a common budget?
09	36/60-month rolling evaluation	When does performance change?
10	Recession and volatility regimes	Where do return and risk differ?
11	Consolidation and combinations	How do the comparisons fit together?
012	Bootstrap uncertainty	How precisely do we estimate performance gaps?

The new contribution is a linked evaluation of architecture and economic conditions.

The completed experiment uses shorter training

Setting	Audited author setup	Our completed run
Stage 1 / 2 / 3 epochs	256 / 64 / 1024	16 / 4 / 48
Optimizer passes	4	4
Checkpoint warm-up	64 epochs	4 epochs
Seeds per architecture	9	9
Training framework	TensorFlow author code	PyTorch port

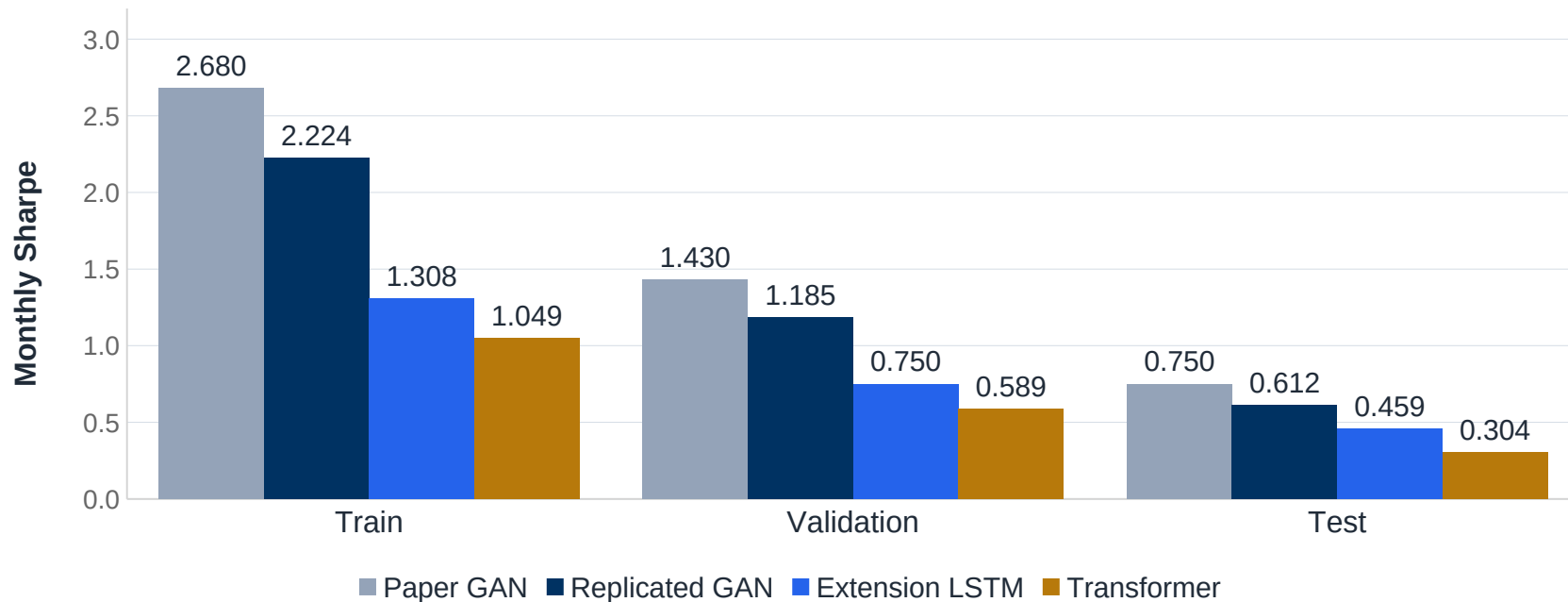
All 18 new models completed in 12.9 minutes. Full training remains a separate test.

LSTM and Transformer encoder comparison

Choice	LSTM	Transformer
Sequence processing	Recurrent state	Causal self-attention
SDF / instrument states	4 / 32	4 / 32
Transformer specification	Not applicable	2 layers, 4 heads, width 32
Economic objective	Shared	Shared
Training seeds and schedule	Shared	Shared

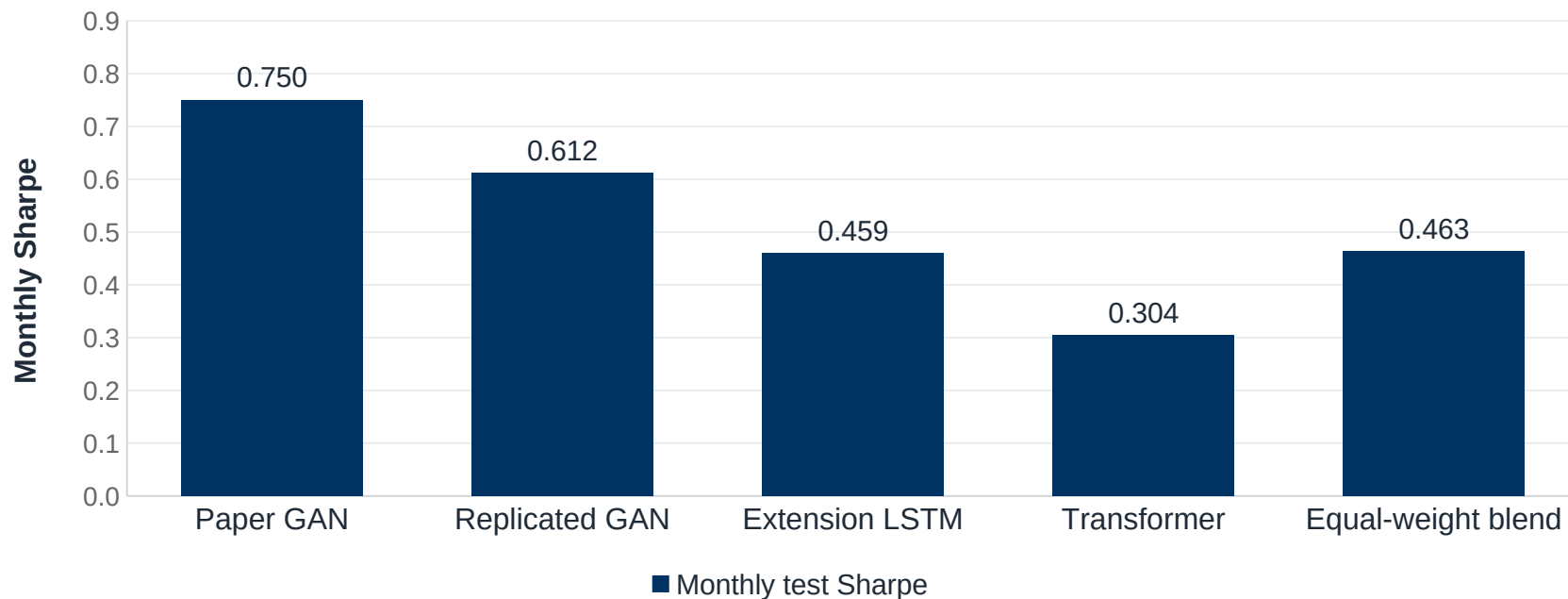
The comparison replaces both macro encoders and keeps the output dimensions.

Train, validation and test comparison



The new LSTM exceeds Transformer in all three splits under the reduced budget.

Full-test Sharpe across the five portfolios



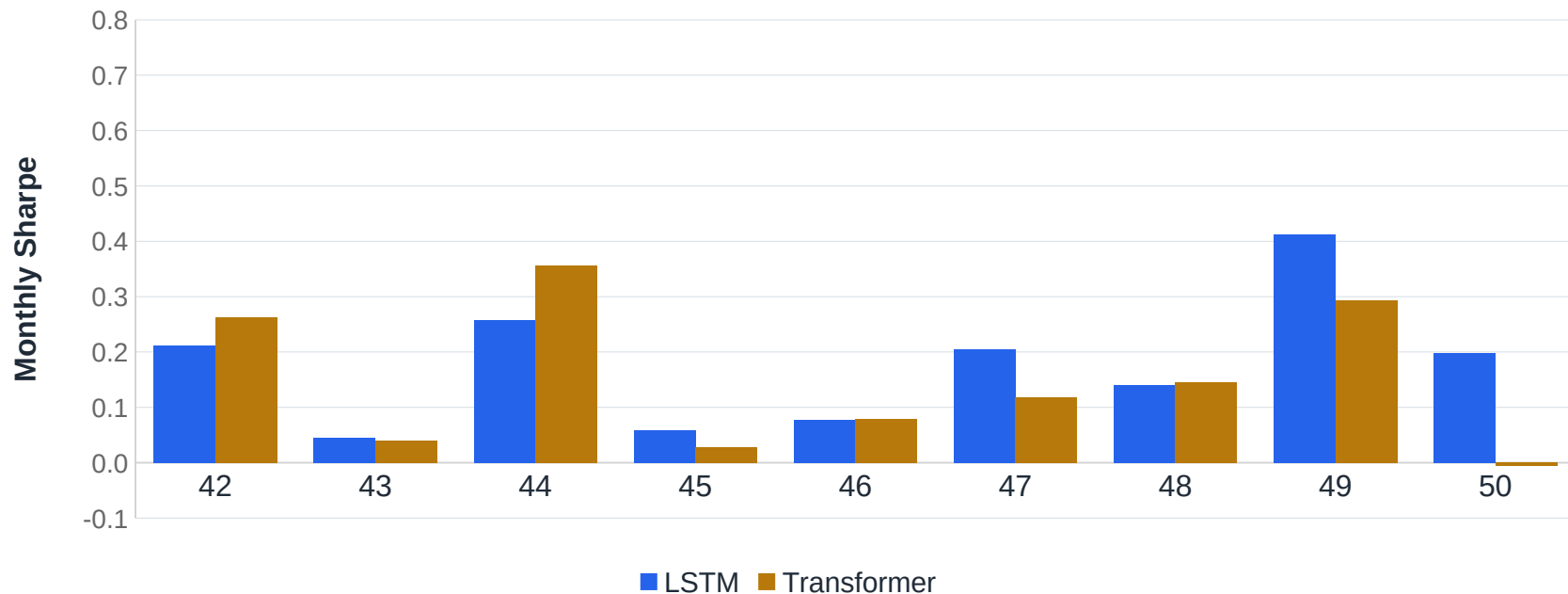
Author GAN 0.750, replication 0.612, extension LSTM 0.459, Transformer 0.304.

Return and risk explain the architecture gap

Portfolio	Mean monthly	Annual volatility	Max drawdown
Paper GAN	0.78%	3.61%	-7.02%
Replicated GAN	0.70%	3.96%	-6.51%
Extension LSTM	0.51%	3.87%	-11.14%
Transformer	0.42%	4.82%	-15.35%
Equal-weight blend	0.47%	3.50%	-6.57%

Transformer combines a lower mean return with greater volatility than new LSTM.

Initialization sensitivity across nine seeds



Ensemble Sharpe uses averaged raw weights, rather than the mean member Sharpe.

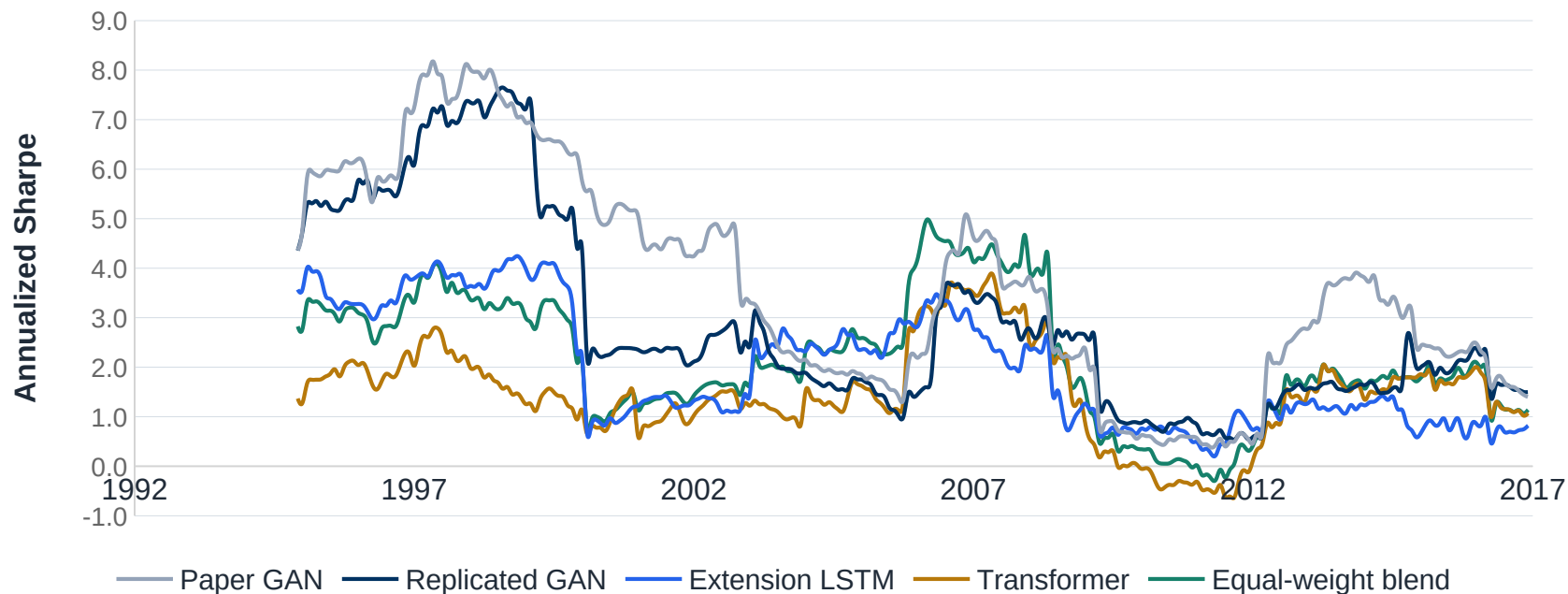
Rolling evaluation uses fixed saved models

Definition	36 months	60 months
Complete windows	265	241
First endpoint	December 1994	December 1996
Last endpoint	December 2016	December 2016
Statistic	Annualized rolling Sharpe	Annualized rolling Sharpe

Each window reuses earlier returns. Longer windows smooth short episodes.

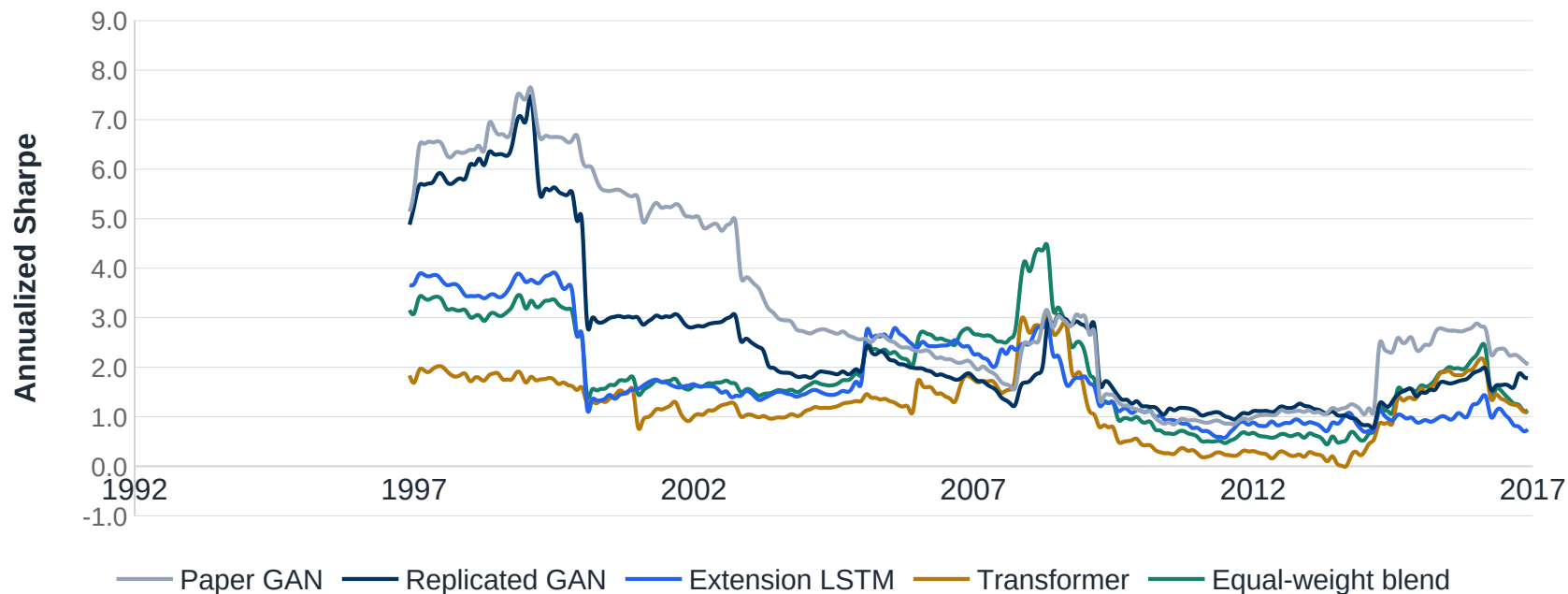
This experiment evaluates fixed models. It does not refit an expanding-window model.

36-month rolling Sharpe



Transformer beats new LSTM in 36.6% of complete windows.

60-month rolling Sharpe



Transformer beats new LSTM in 19.9% of complete windows.

Mean rolling Sharpe comparison

Portfolio	36-month mean	60-month mean
Paper GAN	3.638	3.199
Replicated GAN	2.890	2.504
Extension LSTM	1.992	1.795
Transformer	1.467	1.228
Equal-weight blend	2.166	1.871

LSTM has the higher average, while the 36-month Transformer median is slightly higher.

Economic regime definitions and sample sizes

Regime family	Definition	Test months
Business cycle	NBER expansion / recession	274 / 26
Market volatility	High / low previous-12-month volatility	123 / 177

The volatility cutoff comes from the training sample. Recession dates are retrospective.

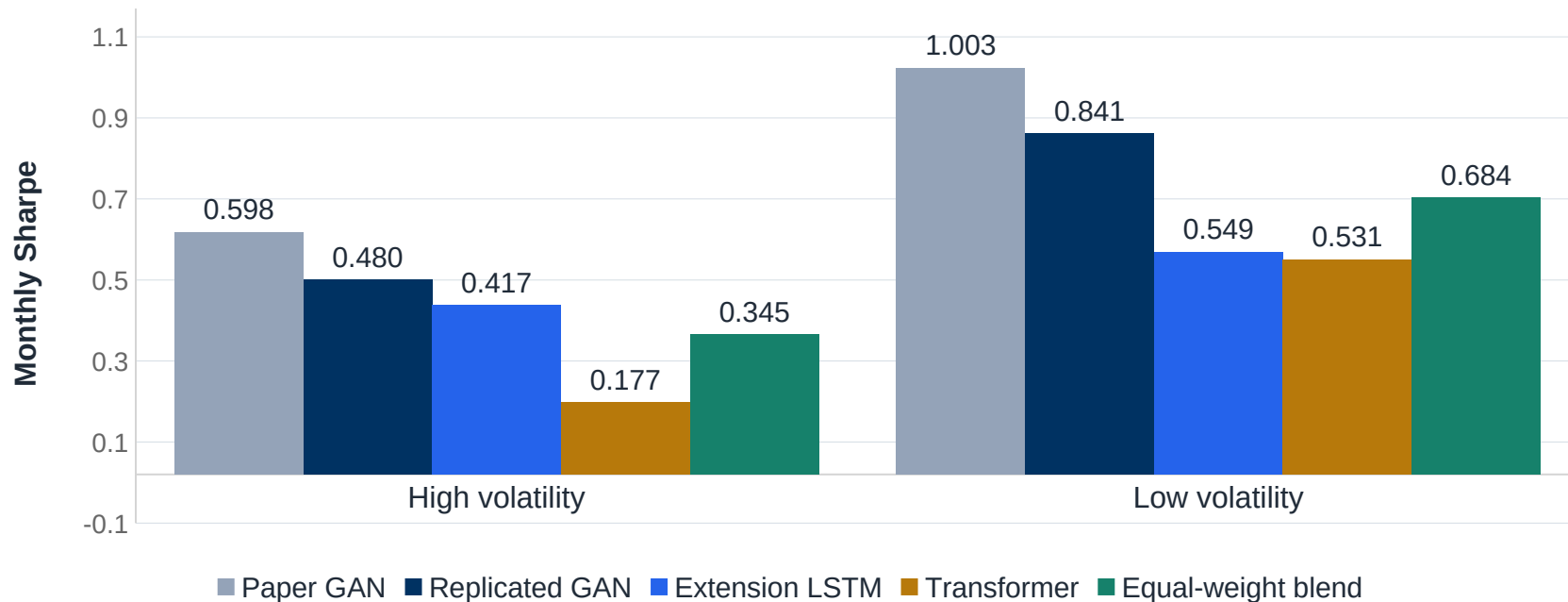
These are separate partitions. Recession estimates depend on only 26 months.

Business-cycle Sharpe comparison



New LSTM outperforms Transformer in both groups. Code 07 exceeds authors GAN in recession.

Market-volatility Sharpe comparison



New LSTM and Transformer are closest in low-volatility months.

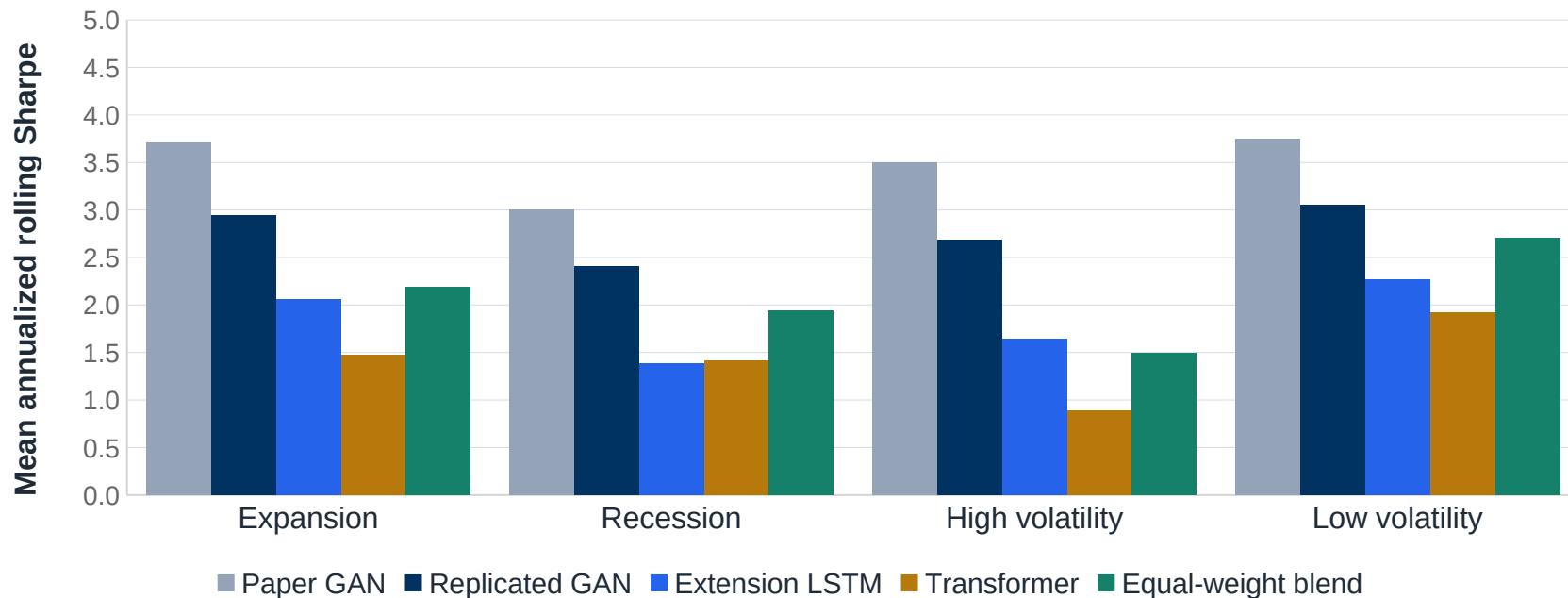
Conditional months and rolling endpoints

Analysis	Returns used	Question
Conditional regime Sharpe	Only labeled months	How does performance differ within the state?
Rolling by endpoint regime	All returns in each trailing window	What trailing performance do we observe in that state?

A 36-month recession-ending window can include many expansion months.

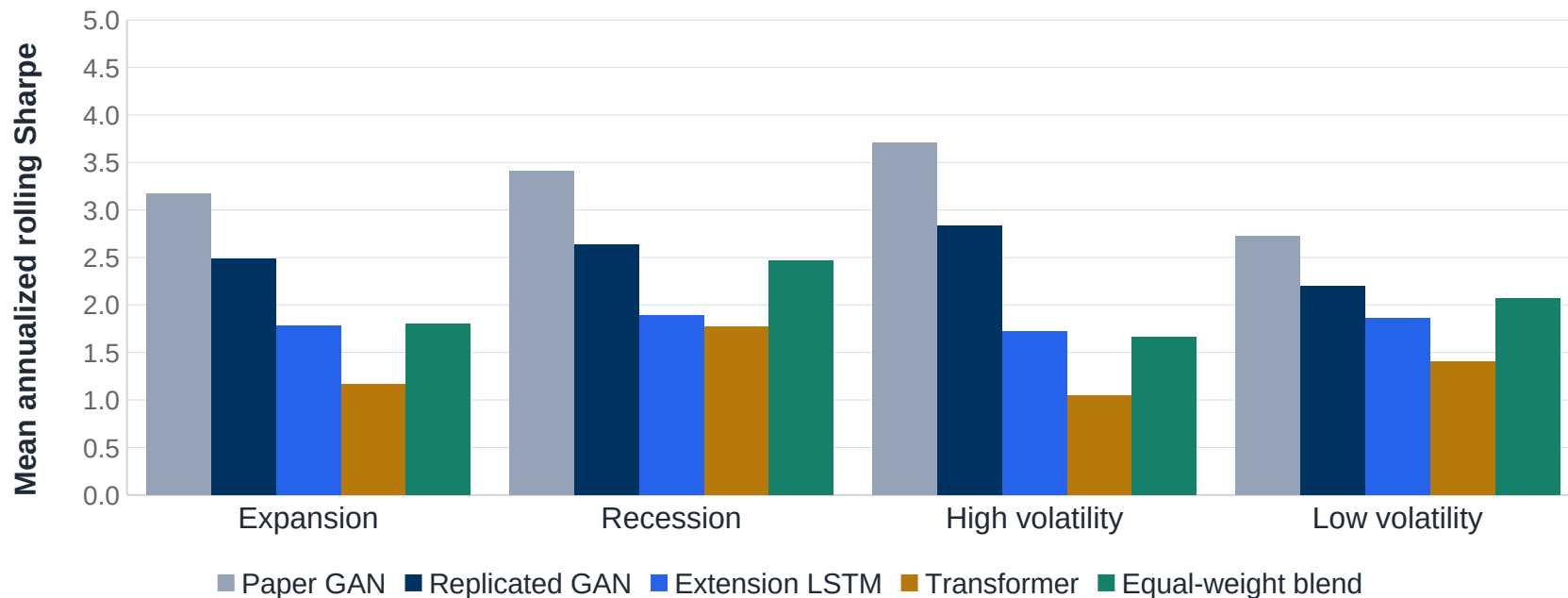
The two measures can rank the models differently because they use different returns.

36-month rolling performance by endpoint regime



In recession-ending windows, Transformer 1.409 slightly exceeds new LSTM 1.383.

60-month rolling performance by endpoint regime



In recession-ending windows, new LSTM 1.893 exceeds Transformer 1.775.

The equal-weight architecture combination

Portfolio	Train	Validation	Test
LSTM	1.308	0.750	0.459
Transformer	1.049	0.589	0.304
50/50 blend	1.414	0.915	0.463

Test correlation between architectures is 0.292. Averaging lowers volatility.

Blend test Sharpe improves only slightly: 0.463 versus LSTM 0.459.

Sampling uncertainty in saved portfolio returns

A paired comparison

Resample the same months for every portfolio, so each draw preserves the return pairing.

Contiguous blocks

Use 10,000 draws and compare 6-, 12- and 24-month blocks to assess dependence sensitivity.

The question

Does a positive observed Sharpe gap remain distinguishable from zero under the bootstrap assumptions?

Uncertainty concerns the return sample, conditional on the saved models.

Full-test model Sharpe and confidence intervals

Portfolio	Monthly Sharpe	95% basic interval
Paper GAN	0.750	[0.442, 0.973]
Replicated GAN	0.612	[0.380, 0.783]
Extension LSTM	0.459	[0.244, 0.626]
Transformer	0.304	[0.146, 0.439]
Equal-weight blend	0.463	[0.265, 0.612]

Individual interval overlap does not answer the paired-difference question.

Paired Sharpe differences and uncertainty

A minus B	Difference	95% basic	95% simultaneous
Paper GAN minus Replicated GAN	0.138	[-0.037, 0.256]	[-0.148, 0.425]
Extension LSTM minus Transformer	0.155	[-0.049, 0.365]	[-0.132, 0.441]
Extension LSTM minus Equal-weight blend	-0.004	[-0.132, 0.119]	[-0.290, 0.283]

A positive point estimate can coexist with an interval containing zero.

LSTM minus Transformer is 0.155, with basic interval spanning zero.

Interval methods can change the conclusion

A minus B	Basic	Percentile	Simultaneous
Paper GAN minus Replicated GAN	Unresolved	A higher	Unresolved
Extension LSTM minus Transformer	Unresolved	Unresolved	Unresolved
Extension LSTM minus Equal-weight blend	Unresolved	Unresolved	Unresolved

“A higher” means the entire interval lies above zero. “Unresolved” means it contains zero.

The paper-versus-replication gap is sensitive to the interval method.

Block-length sensitivity of architecture uncertainty



Both paired basic intervals contain zero at all three block lengths.

Rolling uncertainty changes the interpretation

Rolling window	LSTM point estimate higher	Basic interval favors LSTM	Unresolved
36 months 265 windows	168 / 265	39	226
60 months 241 windows	193 / 241	45	196

No window has a basic interval favoring Transformer. Most windows remain unresolved.

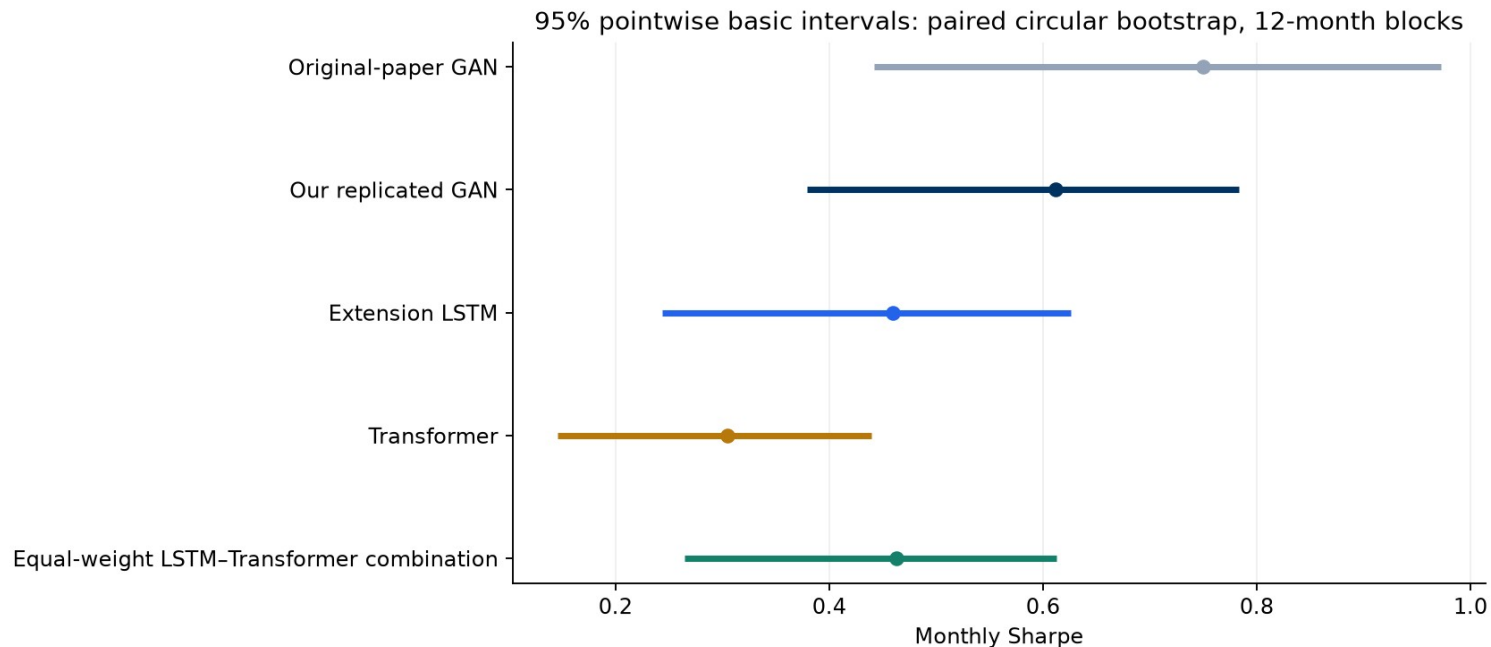
Overlapping window counts describe the history, rather than independent evidence.

Additional comparisons among our own models

Local portfolio	Test Sharpe monthly	36-month mean annualized	Recession Sharpe monthly
Replicated GAN	0.612	2.890	0.455
Feedforward	0.570	2.620	0.570
Equal-weight blend	0.463	2.166	0.076
Extension LSTM	0.459	1.992	0.264
Elastic Net	0.412	2.074	0.254
Linear	0.406	1.740	0.404
Transformer	0.304	1.467	-0.071

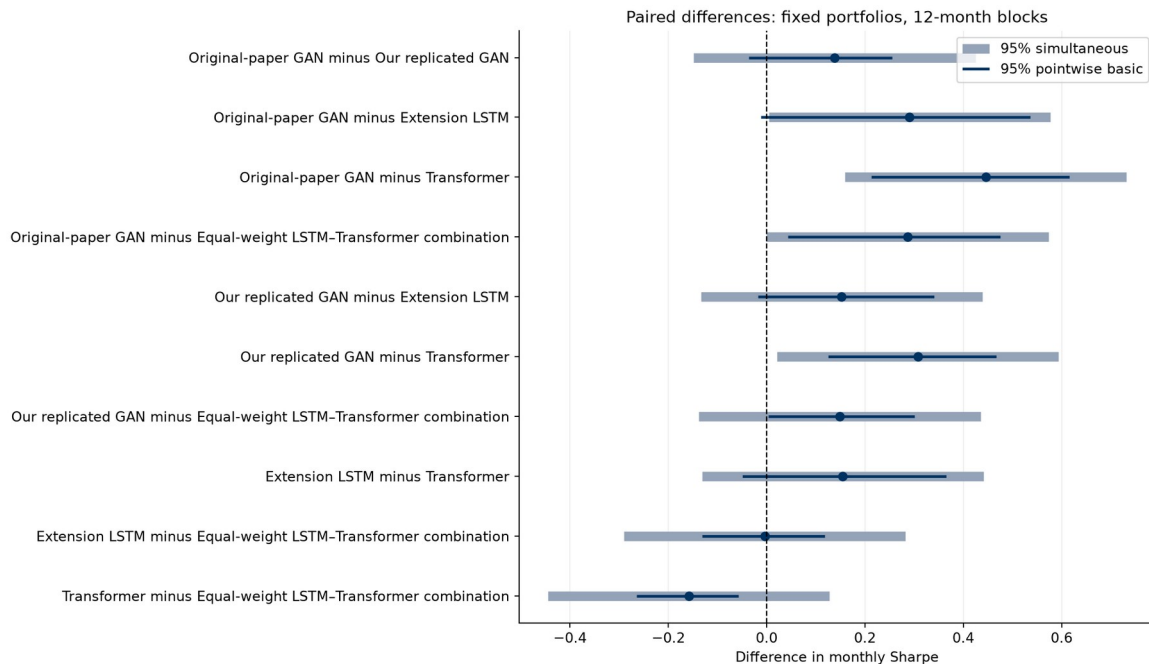
The paper benchmark remains in earlier comparisons. This adds seven local portfolios.

Code 012: model Sharpe confidence intervals



Monthly test Sharpe, with 95% basic intervals using 12-month blocks.

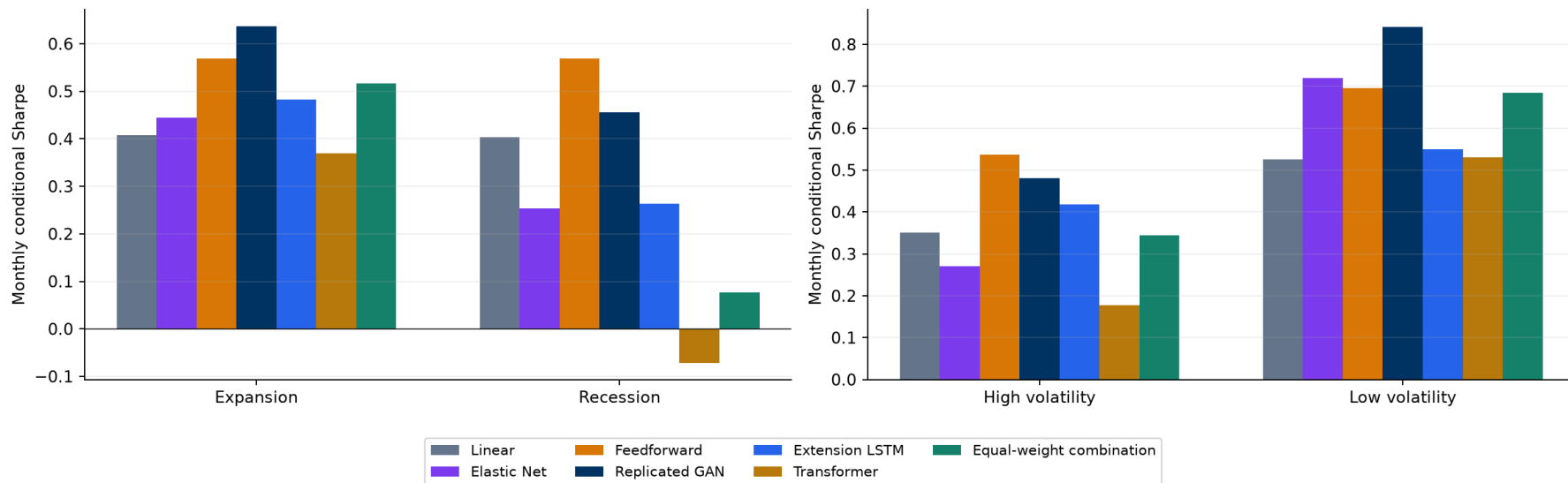
Code 012: uncertainty in paired differences



Paired monthly Sharpe gaps. Pointwise and simultaneous intervals answer different questions.

Code 012: local models across economic regimes

Local models only: business-cycle and volatility comparisons



Conditional monthly Sharpe. These regime estimates do not include confidence intervals.

Contributions and remaining research questions

What we add

A common-budget encoder comparison, rolling and regime evaluation, a fixed combination, and paired uncertainty.

What the evidence supports

LSTM has higher observed test Sharpe. Bootstrap intervals leave its advantage over Transformer unresolved.

What remains open

Full author-schedule GPU training, a new holdout, transaction costs and uncertainty across training runs.

The extensions explain performance differences and the limits of those conclusions.